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This phenomenon, commonly referred to as hallucination, poses significant risks in high-stakes domains such as healthcare, law, finance, and education. In such settings, an incorrect answer can lead to serious consequences, including misdiagnosis, legal misinterpretation, or the spread of misinformation.

Hallucination detection systems are designed to evaluate whether an LLM-generated response is supported by verifiable evidence. These systems typically rely on retrieval-augmented generation pipelines, where relevant documents are retrieved and used as grounding context before or after answer generation.

A core component of hallucination detection is semantic similarity analysis. By embedding both the generated answer and retrieved evidence into a shared vector space, systems can measure how closely the meanings align. Low similarity scores may indicate unsupported claims or fabricated content.

Another critical component is natural language inference (NLI). NLI models assess whether a hypothesis (the generated answer) is entailed by, neutral to, or contradicted by a premise (the evidence). Contradiction detection is particularly valuable, as it can identify responses that directly conflict with source material.

Trust scoring frameworks combine multiple signals, including retrieval overlap, semantic similarity, and contradiction probability, into a single interpretable metric. This score helps users assess the reliability of an AI-generated response at a glance.

Interpretable AI systems emphasize transparency and explainability. Rather than providing a single score, advanced hallucination detection systems surface supporting evidence, highlight unsupported claims, and explicitly flag contradictions. This approach builds user confidence and enables informed decision-making.

As LLMs continue to evolve, evaluation and control mechanisms will become increasingly important. Future research directions include sentence-level hallucination detection, confidence calibration, and benchmarking against standardized datasets such as TruthfulQA and FEVER.

Ultimately, responsible deployment of large language models requires not only powerful generation capabilities but also robust evaluation frameworks that prioritize trust, safety, and accountability.

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